

R7021E : Advanced Robotics and AI

Project Ideas

Control Module

1. Model Predictive Control for a Dynamically Constrained System with a non-trivial body footprint
 - Adopt a nonlinear model of a dynamically constrained center-actuated (or similar) vehicle. An example is an articulated wheel loader model.
 - Modify and extend the MPC lab cost function and constraints.
 - Test a new nonlinear MPC solver. Compare with the solver used in the lab.
 - Add stationary and moving obstacle avoidance constraints.
 - Deploy on a model in Gazebo. If time permits, deploy on a small-scale wheel loader platform that we can provide.

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Navigation and Exploration Module (Extensions of Mobile Robot Exploration Lab)

1. Fully autonomous Multi-Robot Exploration
 - Integrate sensing from multiple robots into one occupancy mapper
 - How do you generate paths for all agents?
 - Frontier evaluation for a multi-robot system – make the connection to task optimization or “travelling salesman” problems.
 - Demonstrate in simulations with multiple robots and if time permits, push for deployment on at least two TurtleBots.
2. Semantic Exploration in Disaster Scenarios
 - Add camera-based detection of objects
 - Extend to use detected objects for exploration guidance (for instance, I see a car, there could be people around it, prioritize exploration there). Implement your own design of information gain while taking semantics of detected objects into account.
 - Set up a realistic simulation world and deploy in simulation.
3. Sampling-based navigation for rapid and dynamically constrained maneuvers:
 - Extend RRT structure into actuation/acceleration or state/pose sampling rather than position-sampling, with constraints like max speed/acceleration
 - Introduce biased sampling, or try different types of sampling methods
 - Analyze the higher dimensional problem (see RRT for manipulators)
 - Extend tree-construction methods such as multi-tree, rrt-connect, etc.
 - Deploy on platform of choice and/or in simulation

Please contact Björn Lindquist (björlin@ltu.se) for further details.

SLAM Module

1. Graph-Based 2D SLAM with Grid Mapping

The student builds a complete pose-graph SLAM system that replaces the particle filter of Lab 4 with a modern front-end/back-end architecture. Laser scans are registered incrementally to produce odometry constraints, revisited places are detected and added as loop-closure constraints, and the resulting trajectory is optimized with a factor-graph solver. The occupancy grid is treated as a pure function of the optimized trajectory and re-rendered whenever the graph changes, so the map is corrected on loop closure instead of accumulating drift.

- Use the scan registration algorithm GICP (small_gicp or Open3D) to align consecutive scans, seeded by wheel odometry, and produce relative-pose constraints with covariances. Keyframes are added on a distance/rotation threshold.
- Radius-based loop closure. Search for keyframes within a radius of the current pose but far apart in time, register the two scans, and accept the match only if the registration fitness passes a gate.
- Factor-graph back-end with GTSAM. Pose2 nodes, BetweenFactorPose2 edges, Huber noise model on loop edges, and iSAM2 for incremental optimization so the map corrects live in RViz.
- Grid mapping from the optimized trajectory. Reuse the lab's GridMap and its inverse sensor model: after each optimization, rebuild the occupancy grid by re-integrating the stored keyframe scans at their corrected poses, and publish it as a standard nav_msgs/OccupancyGrid.
- Evaluate on simulation and hardware runs (either physical robot or rosbags) with at least one real loop. Report ATE and map consistency before and after loop closure, plus runtime per keyframe and per optimization.

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Perception Module

A project related to 'place recognition' during exploration. The project is built on the exploration lab and looks to integrate a camera-enabled computer vision model that does place recognition (covered in the Perception module of the course).

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